Dallas International School

Test 2

Topics 1.8-2.6-2.7

11th IB - Math AA - HL

Name : . . . . . .

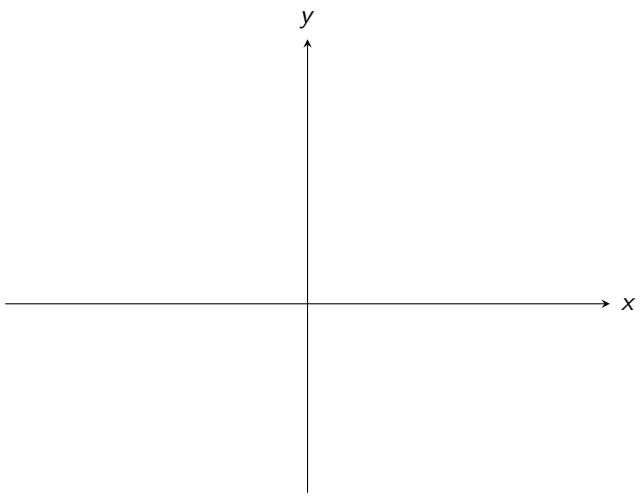
EXERCISE 1 For $y = 3x^2 - 6x - 1$:

1. Write it in the form $y = a(x - h)^2 + k$.

2. State the coordinates of the vertex.

3. Find the y-intercept.

4. Sketch the graph of the quadratic.



EXERCISE 2 The equation $5px^2 + 2px + 1 = p$ has two real, distinct roots.

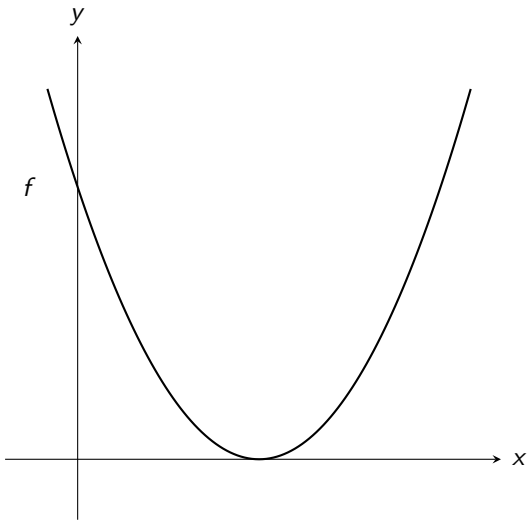
1. Find the possible values for p. [5]

EXERCISE 3

1. Solve $2m^2 - 5m - 3 = 0$.

2. Hence or otherwise, solve $2 \times 9^x - 5 \times 3^x - 3 = 0$.

EXERCISE 4 A function, f, is given by $f(x) = 2x^2 - 12x + p$, where $p \in \mathbb{R}$. The following diagram shows part of the graph of f.



The graph of f has an axis of symmetry $x = q$.

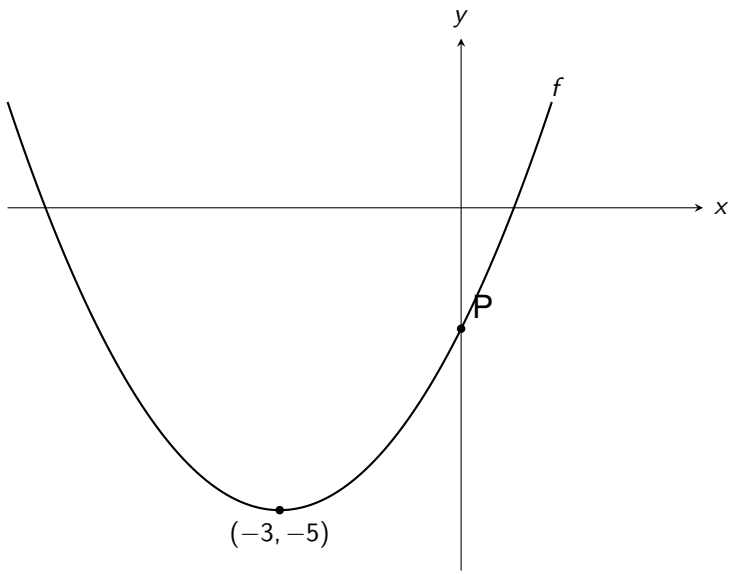
1. Find the value of q.

The vertex of the graph of f lies on the x-axis.

2. a. Write down the value of the discriminant of f.

b. Hence or otherwise, find the value of p.

EXERCISE 5 The following diagram shows part of the graph of a quadratic function f.



The vertex of the parabola is $(-3, -5)$ and the y-intercept is at point P.

1. Write down the equation of the axis of symmetry.

The function can be written in the form $f(x) = \frac{1}{3}(x - h)^2 + k$, where $h, k \in \mathbb{Z}$.

2. a. Write down the values of h and k.

b. Find the y-coordinate of P.